

Amazon ML Challenge 2025: Smart Product Pricing

Team Name: [Your Team Name] **Participant(s):** [Your Name(s)]

1. Overall Methodology

Our approach was to build a **multimodal regression model** that leverages features from product text, engineered numerical data, and image content to predict prices.

We utilized a **hybrid workflow**:

- **Kaggle (GPU Environment)**: Used for the computationally intensive, one-time tasks of downloading the 75,000 product images and extracting deep learning features using a pre-trained model.
- **Local VS Code Environment (CPU)**: Used for all other tasks, including data cleaning, feature engineering, and the rapid training and iteration of our final prediction model.

2. Feature Engineering Techniques

The core of our solution involved robust feature engineering from the raw `catalog_content`.

- **Custom Numerical Features**: We wrote custom Python functions using regular expressions to extract two high-value numerical features:
 - **Item Pack Quantity (IPQ)**: Identified patterns like “(Pack of 6)” or “12 count” to extract the number of items.
 - **Normalized Weight/Volume**: Extracted weight and volume information (e.g., “16 oz”, “5 LB”) and normalized it to a consistent unit (ounces) to make it comparable across products.
 - **Text Cleaning and Vectorization**:
 - The raw text was cleaned by removing HTML tags, converting to lowercase, and stripping punctuation.
 - The cleaned text was then converted into a numerical format using `TfidfVectorizer` on a vocabulary of the 5,000 most important words.
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3. Model Architecture and Algorithms

Our solution employed two distinct models for different tasks:

- **Image Feature Extractor**: `EfficientNetB0`

- We used a state-of-the-art, pre-trained Convolutional Neural Network (CNN) to convert each product image into a 1280-dimension numerical vector (embedding). This allowed us to capture the rich visual information of the products.
 - **Final Price Predictor: LightGBM**
 - We chose a Gradient Boosting Machine (LightGBM) for the final prediction task due to its high performance on tabular data, speed, and excellent handling of sparse feature matrices.
 - To optimize for the competition’s **SMAPE** metric, the model was trained to predict the logarithm of the price (`np.log1p(price)`), which naturally focuses the model on relative (percentage) errors. The final predictions were then converted back to the original scale using `np.expm1()`.
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4. Conclusion

By combining carefully engineered numerical and text features with deep learning-based image features, our LightGBM model was able to effectively learn the complex relationships between product attributes and pricing. This multimodal approach provided a robust solution to the Smart Product Pricing Challenge.